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Submitted

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3h

Peer-graded Assignment: Week 1: Data Cleaning and EDA
30 min

Review Your Peers: Week 1: Data Cleaning and EDA

● Peer-graded Assignment: Week 1: Data Cleaning and EDA

It looks like this is your first peer-graded assignment. [Learn more](#)

You passed!

Congratulations. You earned 25 / 25 points. Review the feedback below and continue the course when you are ready. You can also help more peers by reviewing their submissions.

Review assignments

Instructions

My submission

Discussions

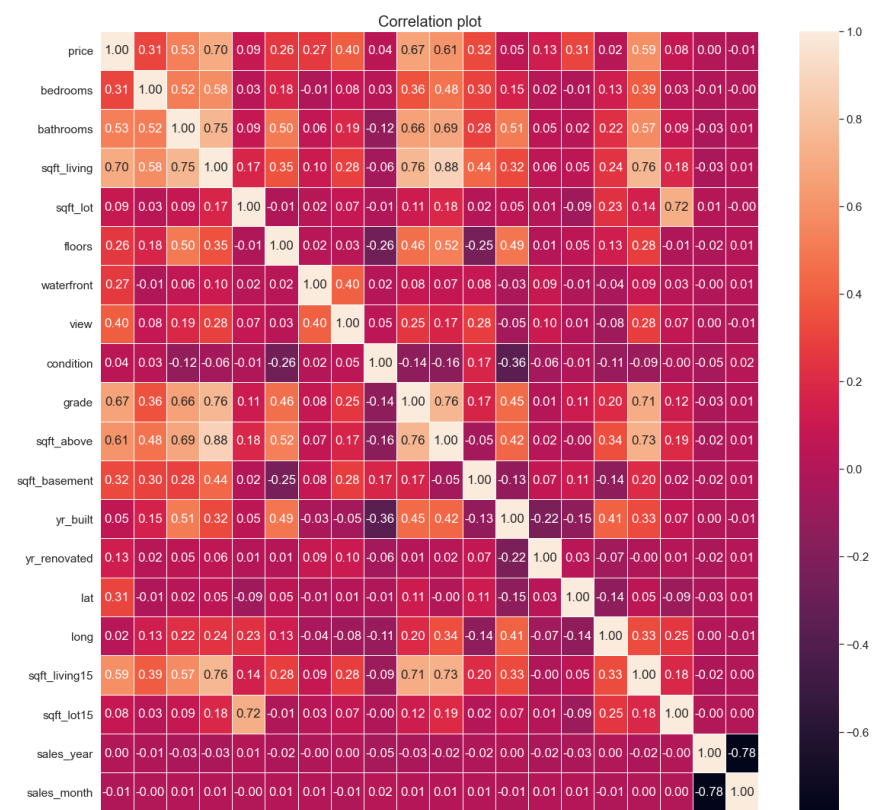
Data Cleaning and EDA Project

Submitted on November 13, 2023

PROMPT

In question 5b of your Jupyter notebook, you **visualized the correlation matrix as a heat map**. Upload a screenshot or copy of your heat map here.

Correlation Plot



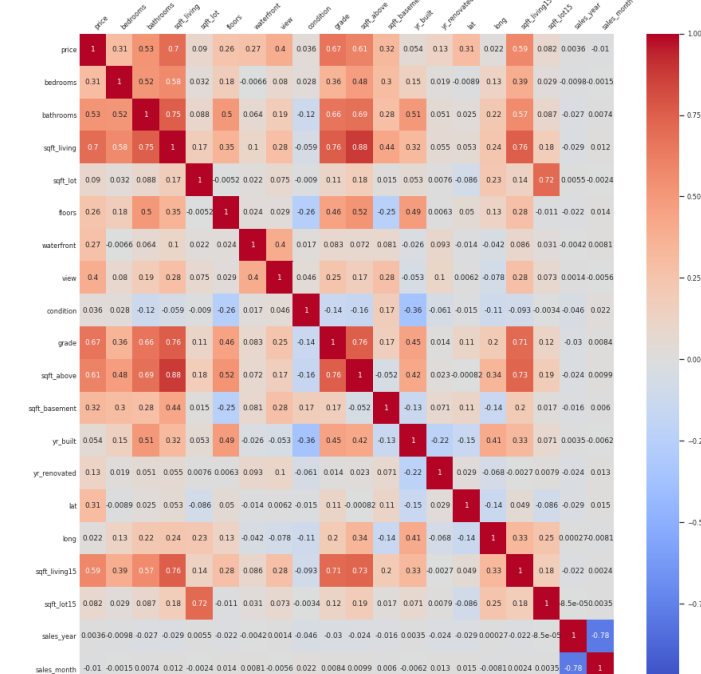
Correlation Plot



RUBRIC

Did the author visualize the correlation matrix as a heat map as directed in question 5b of the lab notebooks?

Here is a sample of visualizing the correlation matrix as a heat map using seaborn:



5 points

The learner visualized the correlation matrix as a heat map, similar to the sample heat map visualization.

3 points

The learner submitted a correlation matrix or some other type of visualization rather than a heat map.

0 points

No visualization submitted.

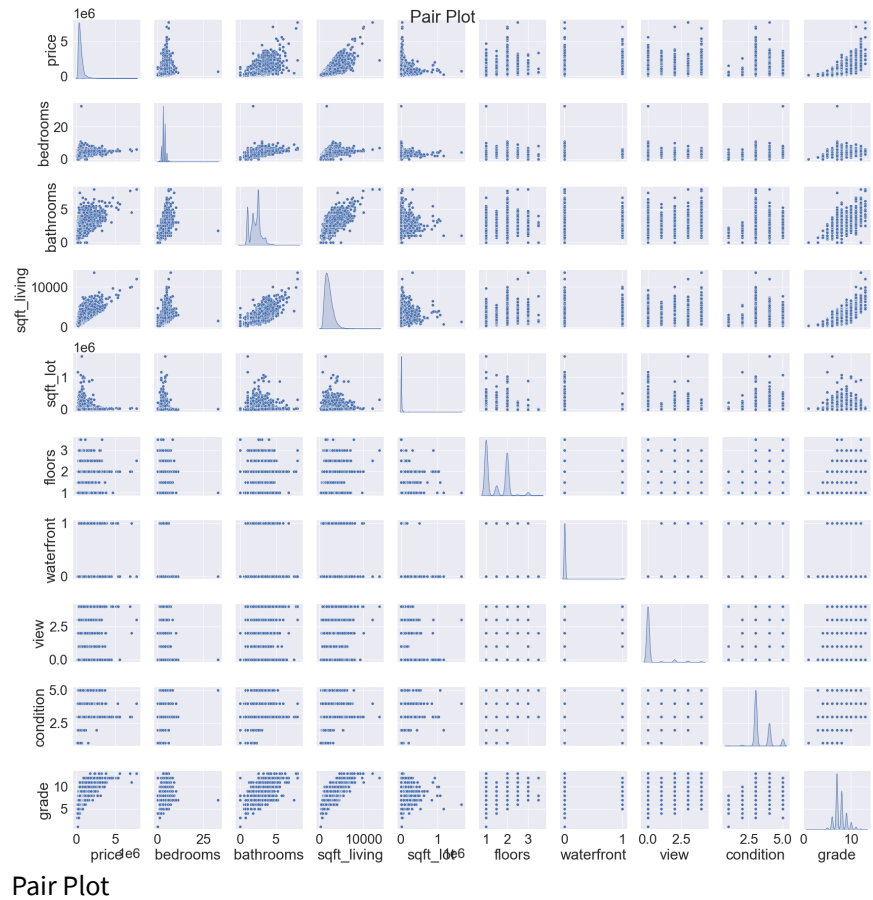
PROMPT

The directions for 5c in your Jupyter notebook are as follows:

Pair plot is a fast way to inspect relationships between features. Use seaborn's .pairplot() function to draw a pair plot of the first 10 columns (including price) and inspect their relationships. Set the diagonal elements to be KDE plot.

Upload a copy or a screenshot of your pair plot.

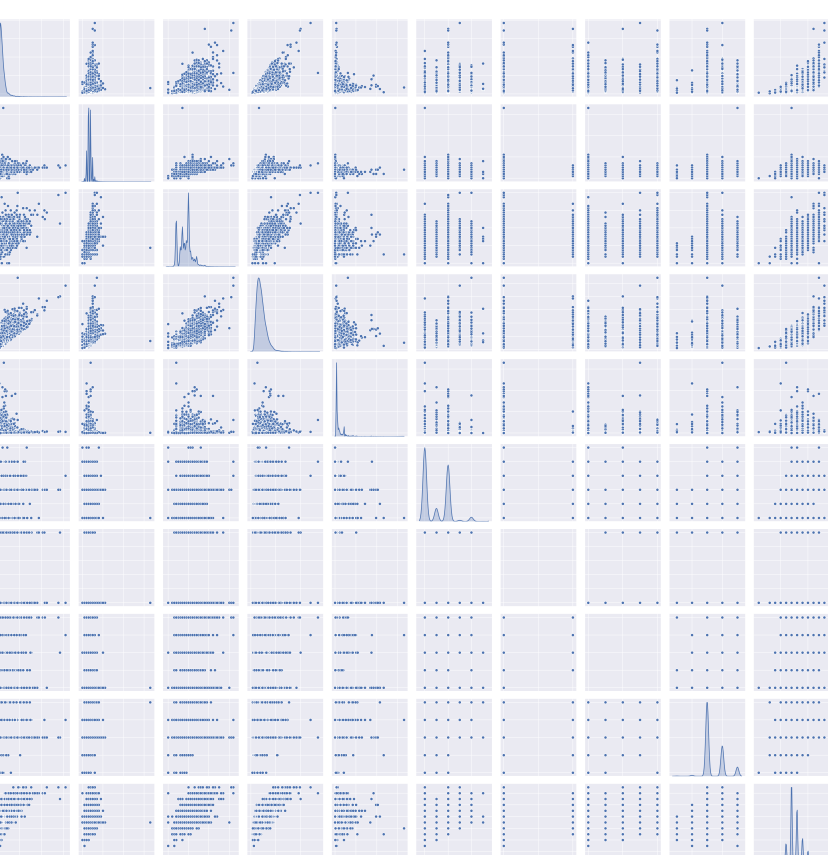
Pair Plot



RUBRIC

Evaluate the learner's pair plot solution.

Here is a **sample pair plot solution**:



5 points

Pair plot **includes both of the following**: the first ten columns (including price) **and** the diagonal elements set to KDE plot.

3 points

Pair plot **missing at least one of the following**: includes first ten columns (including price) **and** diagonal elements set to KDE plot.

0 points

No submission or submitted a different type of plot than a pair plot.

PROMPT

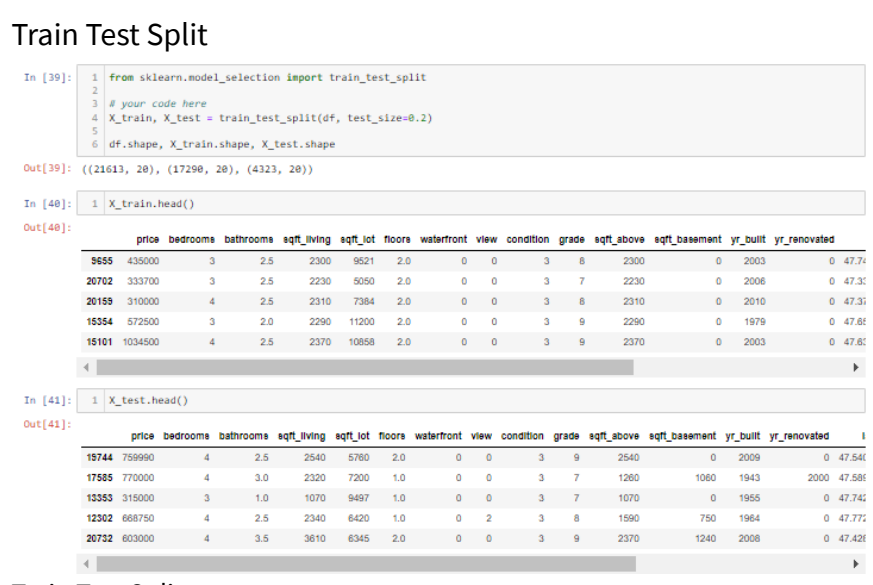
The directions for 6a are:

We will split the data to train and test datasets such that the test dataset is 20% of original data. Use the **sklearn.model_selection.train_test_split** function to split the data frame to X_train and X_test. X_train is 80% of observation randomly chosen. X_test is the rest 20%.

Both X_train and X_test are **pd.DataFrame** object and include 'price' in the table. Note that the train_test_split can handle data frame as well as array.

Upload a screenshot of your code for splitting the data frame into training and testing sets.

Train Test Split



RUBRIC

Here is the code from the solution:

```
1 from sklearn.model_selection import train_test_split
2 X_train, X_test = train_test_split(df2, test_size=0.2, random_state=42)
3 print(len(X_train), len(X_test))
```

Evaluate the learner's train-test split.

5 points

Includes all of the following: uses the **sklearn.model_selection.train_test_split** function and appears to pass all original data to the function and creates X_train and X_test and test_size=0.2 so 20% of the original set is the test set and 80% is the training set.

3 points

Uses **sklearn.model_selection.train_test_split** function but **missing one of the following**: appears to pass all original data to the function **or** creates X_train and X_test **or** test_size=0.2 so 20% of the original set is the test set and 80% is the training set.

1 point

Uses **sklearn.model_selection.train_test_split** function but **missing at least two of the following**: appears to pass all original data to the function **or** creates X_train and X_test **or** test_size=0.2 so 20% of the original set is the test set and 80% is the training set.

0 points

Does not use sklearn.model_selection.train_test_split function.

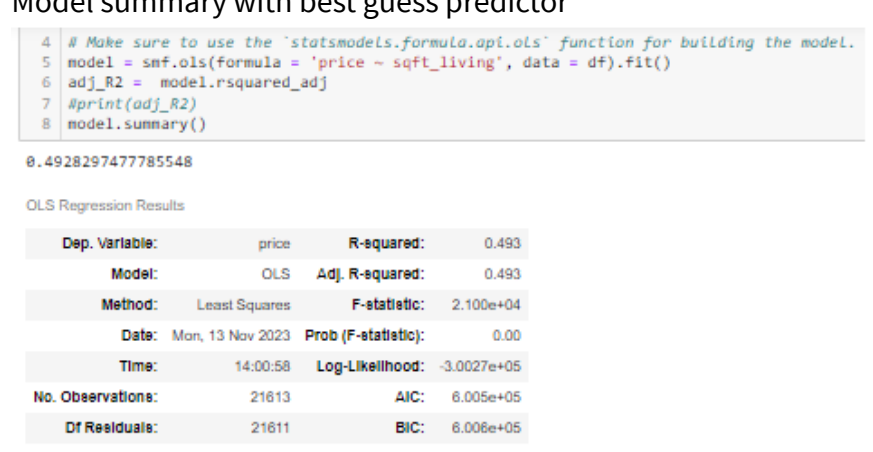
PROMPT

The directions for 6b are:

Use the **best_guess_predictor** as a single predictor and build a simple linear regression model using the **statsmodels.formula.api.ols** function (https://www.statsmodels.org/dev/example_formulas.html). Train on the X_train portion. Print out the results summary.

Upload a screenshot or copy of the results summary here.

Model summary with best guess predictor



RUBRIC

Evaluate the learner's results summary.

Here is the **results summary from the solution**:

```
===== OLS Regression Results =====
Dep. Variable: price                R-squared: 0.485
Model: OLS                        Adjusted R-squared: 0.479
Method: Least Squares              F-statistic: 1.87e+02
Date: Mon, 13 Nov 2023              Prob(>F): 0.000
Time: 14.0508                      Log Likelihood: -3.0027e+02
No. Observations: 2103              AIC: 6.005e+02
Df Residuals: 2101                  BIC: 6.030e+02
Date: Mon, 13 Nov 2023              Prob(>F): 0.000
Time: 14.0508                      Log Likelihood: -3.0027e+02
No. Observations: 2103              AIC: 6.005e+02
Df Residuals: 2101                  BIC: 6.030e+02
```

Regression Results

	price	R-squared
Model:	OLS	0.485
Method:	Least Squares	0.485
Date:	Mon, 13 Nov 2023	Prob(>F): 0.000
Time:	14.0508	Log Likelihood: -3.0027e+02
No. Observations:	2103	AIC: 6.005e+02
Df Residuals:	2101	BIC: 6.030e+02

Regression Results

(1) Standard errors assume that the covariance matrix of the errors is correctly specified.
(2) The coefficient values of intercept, slope, and error term indicate that there are strong multicollinearity in other numerical predictors.

